STANDATA interpretation 23-BCI-015

Building

Wood roof truss systems in Part 9 buildings

Date Issued: February 2025

Purpose

To provide clarity of the requirement for professional involvement in wood roof truss system designs used in Part 9 buildings.

Discussion

Wood roof truss systems have become the preferred construction method for roofs of residential and small commercial buildings falling under the scope of Part 9 of the National Building Code – 2023 Alberta Edition (NBC(AE)). These systems are customizable and can accommodate many exterior wall configurations. Due to their customizability and complexity designers with specific knowledge are required to design the wood roof trusses.

Article 9.23.14.11. of the NBC(AE) refers to "good engineering practice" for the design of wood roof trusses and refers to the requirements in Subsection 4.3.1. for the design of joint connections used in any roof truss, respectively. The term "good engineering practice" requires the involvement of an individual who is authorized to engage in the practice of engineering under the *Engineering and Geoscience Professions Act* and its Regulations. Designs carried out in accordance with Part 4 of the NBC(AE) require the designer be a registered professional skilled in the work concerned.

Based on the requirements of Article 9.23.14.11., a wood roof truss system must be designed by a registered engineering professional, including the way trusses are connected to other trusses. In addition to the design inputs for and details of individual trusses within the system, sufficient detail of how trusses are interconnected is the responsibility of the registered engineering professional to design and authenticate. Authentication does not have to occur on the roof truss layout plan provided the information of how trusses are interconnected is provided in a separate authenticated document. It is up to the registered engineering professional as to how they choose to communicate their design and authenticate it. Authorities having jurisdiction should review any exclusions to the authentication noted on the drawings and ensure any limitations are appropriate.

The Permit Regulation provides for terms and conditions to be imposed on the permit to facilitate the processing of the application for issuance. Authenticated truss designs may not be available at the time of permit application due to the procurement process. In such cases, the submission of the authenticated truss drawings may be a condition of the permit for submission prior to the request for a framing inspection. In any case the approval of a safety codes officer is required before any part of the building or system is covered or concealed.

Code References

Division A, Article 1.4.1.2.

1.4.1.2. Defined Terms

1) The words and terms in italics in this Code shall have the following meanings:

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Designer means the person responsible for the design.

Licensed interior designer means an individual who is authorized to engage in that portion of the practice of architecture that is defined as interior design under the Architects Act and its Regulations.

Registered architectural professional means an individual who is authorized to engage in the practice of architecture under the Architects Act and its Regulations.

Registered engineering professional means an individual who is authorized to engage in the practice of engineering under the Engineering and Geoscience Professions Act and its Regulations.

Registered professional means an individual who qualifies as a
(a) *registered architectural professional*,
(b) *registered engineering professional*, or
(c) *licensed interior designer*.

Article 9.23.14.11.

9.23.14.11. Roof Trusses

1) Wood roof trusses shall be designed in accordance with good engineering practice such as that described in TPIC 2019, "Truss Design Procedures and Specifications for Light Metal Plate Connected Wood Trusses."

2) The joint connections used in trusses described in Sentence (1) shall be designed in conformance with the requirements in Subsection 4.3.1. (See Note A-9.23.14.11.(2).)

3) All member bracing shall be installed as per the truss design drawings, and continuous lateral bracing shall be adequately anchored to the roof and ceiling diaphragms at intervals no greater than 6.10 m o.c.

A-9.23.14.11.(2) Wood Roof Truss Connections. Sentence 9.23.14.11.(2) requires that the connections used in wood roof trusses be designed in conformance with Subsection 4.3.1., and Sentence 2.4.1.2.(1) of Division C, which applies to all of Part 4, requires that the designer be a registered professional skilled in the work concerned. This has the effect of requiring that the trusses themselves be designed by registered professionals. Although this is a departure from the usual practice in Part 9, it is appropriate, since wood roof trusses are complex structures which depend on a number of components (chord members, web members, cross-bracing, connectors) working together to function safely. This complexity precludes the standardization of truss design into tables comprehensive enough to satisfy the variety of roof designs required by the housing industry.

Division C, Sentence 2.4.1.2.(1)

2.4.1.2. Structural Design

1) For design carried out in accordance with Part 4 of Division B, the *designer* shall be a *registered professional* skilled in the *work* concerned. (See Note A-2.4.1.2.(1).)

A-2.4.1.2.(1) Structural Design. Part 4 of Division B is written on the assumption that structural design will be carried out by a registered professional who is qualified to perform such design. Sentence 2.4.1.2.(1) is not intended to imply that a registered professional may not also be required in the application of requirements in other Parts of the NBC(AE).

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Permit Regulation

Terms and conditions of permit

22 A permit issuer may issue a permit for an undertaking, or part of the undertaking, and may, without limiting the generality of section 44(2) of the Act and subject to the applicable policies of the Minister and of the accredited municipality or accredited regional services commission, impose terms and conditions on the permit that are consistent with the purpose and intent of the Regulation, including, but not limited to,

(a) requiring that permission be obtained from the permit issuer before the occupancy or use of the construction, process or activity under the permit,
(b) setting the date on which the permit expires,
(c) setting a condition that causes the permit to expire,
(d) setting the period of time that the undertaking may be occupied, used or operated,
(e) setting out the scope of the undertaking being permitted,
(f) setting the location or locations of the undertaking being permitted,
(g) setting the qualifications of the person responsible for the undertaking,
(h) requiring an identification number or label to be affixed to the undertaking, and
(i) requiring the approval of a safety codes officer before any part of the building or system is covered or concealed.

Application

This interpretation applies to the design of all wood roof truss systems used in Part 9 buildings.

Interpretation

1. The design of a wood roof truss system must be designed and authenticated by a registered engineering professional.

2. Wood roof truss system layouts are required to be authenticated unless the essential information needed to design and install the roof trusses is provided in a separate document that is already authenticated. This could be the truss profile drawings or the framing plan of the building drawings.

3. The authenticated truss profile drawings or other required authenticated drawings may be provided to the authority having jurisdiction prior to their approval for covering or permission for occupancy as part of the permit conditions.

This INTERPRETATION is applicable throughout the Province of Alberta.

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Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code – 2023 Alberta Edition

Issued by the Provincial Building Administrator

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